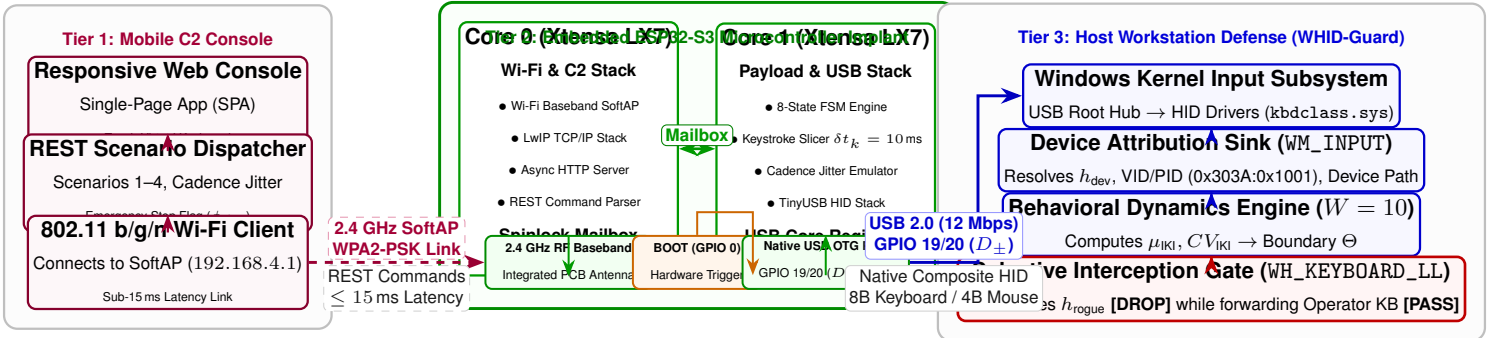

Tier 1: Mobile C2 Console
Responsive Web Console
Single-Page App (SPA)
REST Scenario Dispatcher
Scenarios 1–4, Cadence Jitter
802.11 b/g/n Wi-Fi Client
Connects to SoftAP (192.168.4.1)
Sub-15 ms Latency Link
2.4 GHz SoftAP
WPA2-PSK Link
REST Commands
≤ 15 ms Latency
Core 0 (Xtensa LX7)
Wi-Fi & C2 Stack
Wi-Fi Baseband SoftAP
LwIP TCP/IP Stack
Async HTTP Server
REST Command Parser
Spinlock Mailbox
2.4 GHz RF Baseband
Integrated PCB Antenna
Mailbox
Core 1 (Xtensa LX7)
Payload & USB Stack
8-State FSM Engine
Keystroke Slicer $\delta t_k = 10$ ms
Cadence Jitter Emulator
TinyUSB HID Stack
BOOT (GPIO 0)
Hardware Trigger
Native USB OTG
GPIO 19/20 ($D_\pm$)
USB 2.0 (12 Mbps)
GPIO 19/20 ($D_\pm$)
Native Composite HID
8B Keyboard / 4B Mouse
Tier 3: Host Workstation Defense (WHID-Guard)
Windows Kernel Input Subsystem
USB Root Hub → HID Drivers (kbdclass.sys)
Device Attribution Sink (WM_INPUT)
Resolves h_{dev}, VID/PID (0x303A:0x1001), Device Path
Behavioral Dynamics Engine ($W = 10$)
Computes μ_{IKI}, CV_{IKI} → Boundary Θ
Active Interception Gate (WH_KEYBOARD_LL)
h_{rogue} [DROP] while forwarding Operator KB [PASS]